Performance of a Network

- The performance of a network pertains to the measure of service quality of a network as perceived by the user.

- There are different ways to measure the performance of a network, depending upon the nature and design of the network.

- Finding the performance of a network depends on both **quality** of the network and the **quantity** of the network.

Parameters for Measuring Network Performance

- Bandwidth
- Latency (Delay)
- Bandwidth - Delay Product
- Throughput
- Jitter

BANDWIDTH

- One of the most essential conditions of a website's performance is the amount of bandwidth allocated to the network.
- Bandwidth determines how rapidly the webserver is able to upload the requested information.
- Bandwidth is characterized as the measure of data or information that can be transmitted in a fixed measure of time.
- In the case of **digital devices**, the bandwidth is measured in **bits per second(bps)** or **bytes per second**. In the case **of analog devices**, the bandwidth is measured in **cycles per second** or **Hertz (Hz).**

- People frequently mistake bandwidth with internet speed in light of the fact that Internet Service Providers (ISPs) tend to claim that they have a fast "**40Mbps connection**" in their advertising campaigns.

- True internet speed is actually the amount of data you receive every second and that has a lot to do with latency too.

"Bandwidth" means **"Capacity"** and **"Speed"** means **"Transfer rate".**

- More bandwidth does not mean more speed

LATENCY

- In a network, during the process of data communication, **latency(also known as delay)** is defined as the total time taken for a complete message to arrive at the destination, starting with the time when the first bit of the message is sent out from the source and ending with the time when the last bit of the message is delivered at the destination.

- High latency leads to the creation of bottlenecks in any network communication- conclusively decreases the **bandwidth**

- Latency is also known as a **ping rate** and is measured in **milliseconds(ms).**

- ***Propagation time = Distance / Propagation speed***

- What will be the propagation time when the distance between two points is 12, 000 km?

Assuming the propagation speed to be 2.4 * 10^8 m/s in cable.

- Propagation time = (12000 * 1000) / (2.4 * 10^8) = 0.05s=50 ms

- **Transmission time = Message size / Bandwidth**

What will be the propagation time and the transmission time for a 2.5-kbyte message when the bandwidth of the network is 1 Gbps?

Assuming the distance between sender and receiver is 12, 000 km and speed of light is 2.4 * 10^8 m/s.

Propagation time = (12000 * 1000) / (2.4 * 10^8) = 50 ms

Transmission time = (2560 * 8) / 10^9 = 0.020 ms

BANDWIDTH - DELAY PRODUCT

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The **Bandwidth-Delay Product (BDP)** tells us **how much data can be "in flight" on a network at one time**. "In flight" means the data has been sent but has not yet reached the receiver**not yet reached the receiver**.

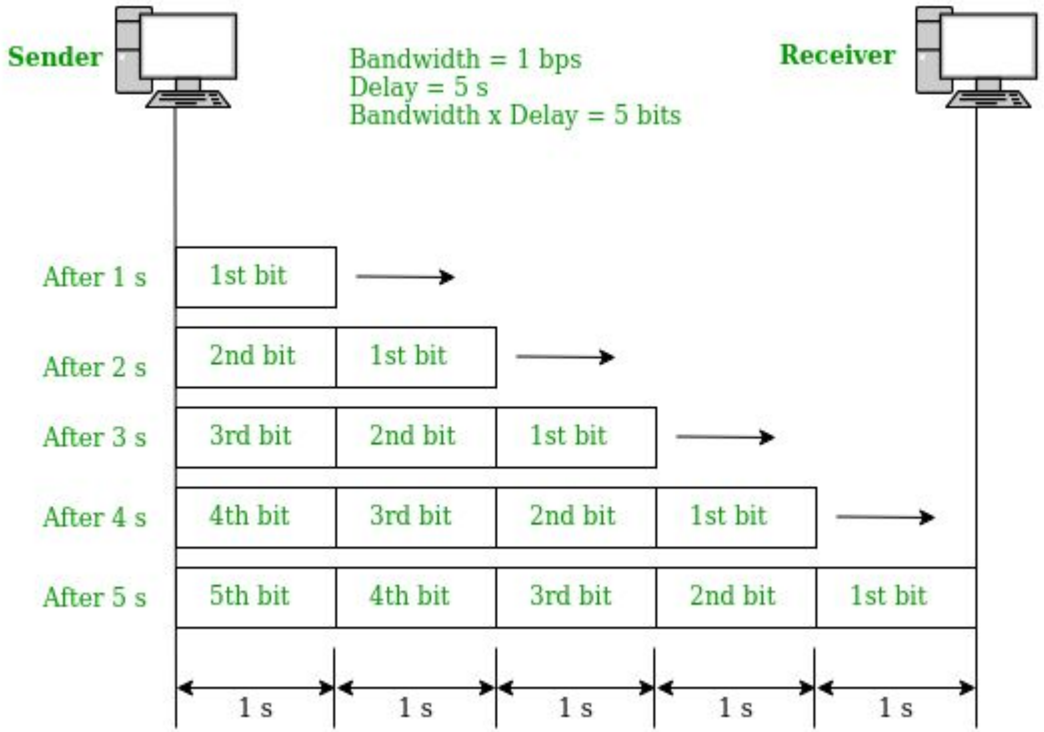
$$\boxed{\text{Bandwidth–Delay } Product = Bandwidth \times Propagation\ Delay}$$

Bandwidth = How many bits can be sent every second (bits/s).

Delay (Propagation Delay) = Time taken by a bit to travel from sender to receiver (seconds).

The result is measured in **bits**.

- **Case 1:** Assume a link is of bandwidth 1bps and the delay of the link is 5s. Let us find the bandwidth-delay product in this case.

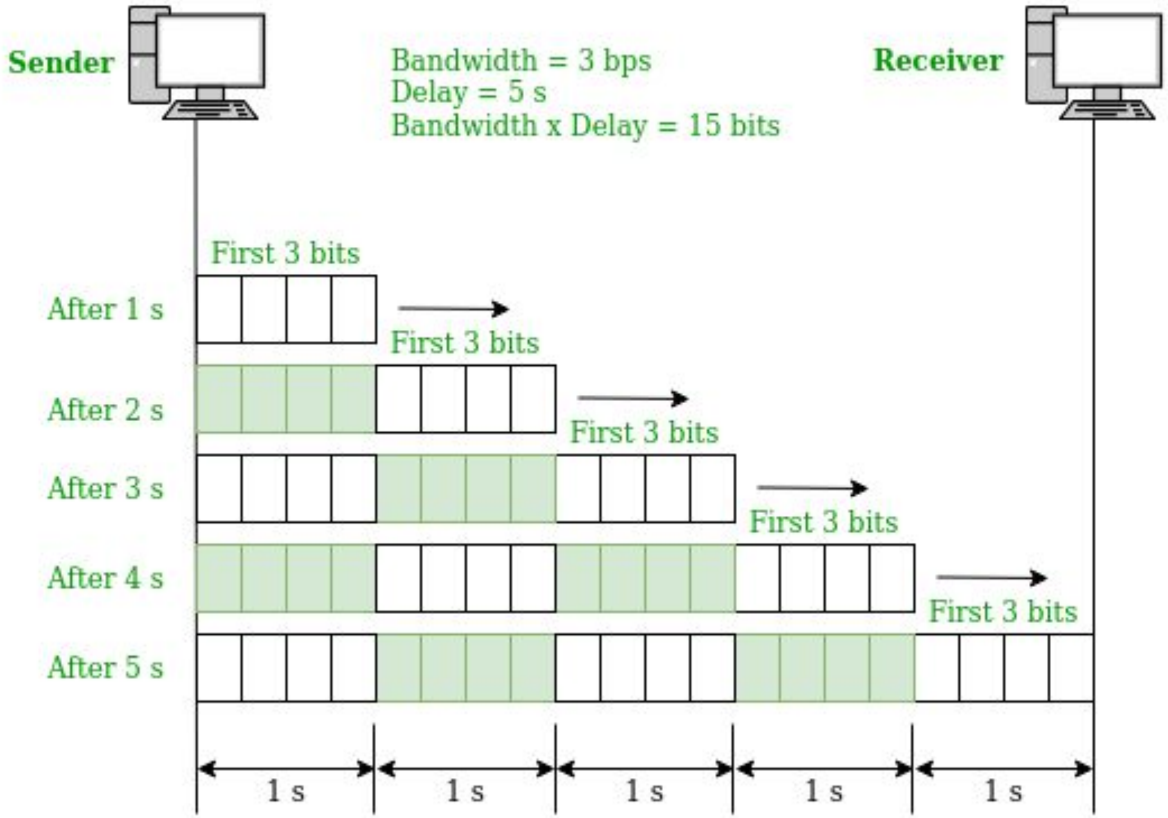


- The sender sends **1 bit every second**.
- The first bit takes **5 seconds** to reach the receiver.
- Before the first bit arrives, the sender has already sent:

Time	Bits Sent	Bits Still on Link
1 s	1	1
2 s	2	2
3 s	3	3
4 s	4	4
5 s	5	5

- After 5 seconds, the first bit reaches the receiver.
- So, **at most 5 bits are present on the link simultaneously.**
- **BDP = 5 bits**

- **Case 2:** Assume a link is of bandwidth 3bps.



- Now the sender sends **3 bits every second**.

Time	Bits Sent	Bits on Link
1 s	3	3
2 s	6	6
3 s	9	9
4 s	12	12
5 s	15	15

- After 5 seconds, the first 3 bits reach the receiver.
- So, **15 bits can be in transit at the same time.**
- **BDP = 15 bits**

Why is Bandwidth-Delay Product Important?

- Suppose a sender transmits data and **waits for an acknowledgment (ACK)** before sending more.
- If the sender sends **too little data**, the network stays idle while waiting for the ACK, wasting bandwidth.

Bandwidth = **100 Mbps**

Delay = **0.05 s (50 ms)**

Then,

$$\text{BDP} = 100 \times 10^6 \times 0.05 = 5{,}000{,}000\ bits$$

- This means **5 million bits can already be traveling through the network** before the first bit arrives.

- If the sender sends only **1000 bits** and waits for an ACK, the network remains mostly empty, resulting in poor performance.

Why Use $2 \times Bandwidth \times Delay$

In a **full-duplex** network:

- Data travels **from sender to receiver**.
- ACKs travel **from receiver back to sender**.

The sender waits for the ACK, which takes another propagation delay to return.

So the total waiting time is approximately the **Round-Trip Time (RTT)**:

$$RTT \approx 2 \times Propagation\ Delay$$

THROUGHPUT

- Throughput is the number of messages successfully transmitted per unit time. It is controlled by available bandwidth, the available signal-to-noise ratio, and hardware limitations.

- The maximum throughput of a network may be consequently higher than the actual throughput achieved in everyday consumption.

- **Throughput** is an actual measurement of **how fast we can send data**.

A network with bandwidth of 10 Mbps can pass only an average of 12, 000 frames per minute where each frame carries an average of 10, 000 bits. What will be the throughput for this network?

- We can calculate the throughput as-

  Throughput = (12, 000 x 10, 000) / 60 = 2 Mbps

- The throughput is nearly equal to one-fifth of the bandwidth in this case.

JITTER

- Jitter is another performance issue related to the delay. In technical terms, **jitter is a "packet delay variance".**

- It can simply mean that jitter is considered a problem when different packets of data face different delays in a network and the data at the receiver application is time-sensitive, i.e. audio or video data.

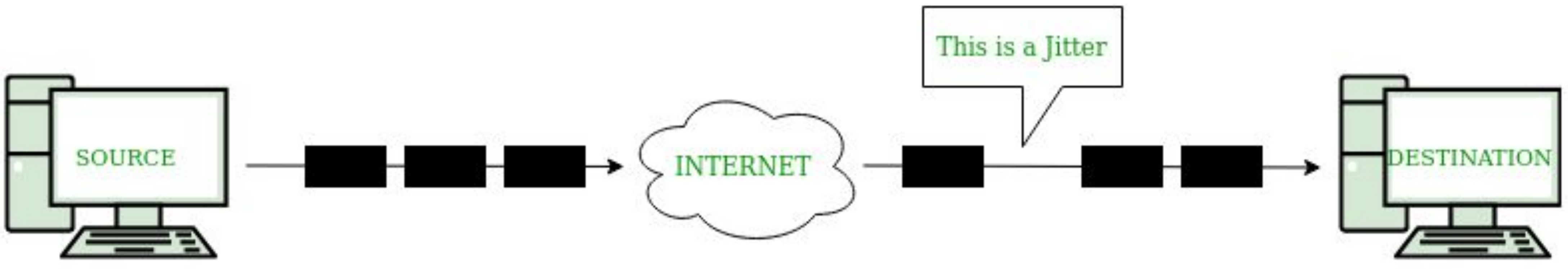
- **Jitter is measured in milliseconds(ms).**

Jitter is harmful and causes network congestion and packet loss.

- **Congestion** is like a traffic jam on the highway. Cars cannot move forward at a reasonable speed in a traffic jam. Like a traffic jam, in congestion, all the packets come to a junction at the same time. Nothing can get loaded.

- **Packet loss :** When packets arrive at unexpected intervals, the receiving system is not able to process the information, which leads to missing information also called "packet loss".

- This has negative effects on video viewing. If a video becomes pixelated and is skipping, the network is experiencing a jitter. The result of the jitter is packet loss. When you are playing a game online, the effect of packet loss can be that a player begins moving around on the screen randomly. Even worse, the game goes from one scene to the next, skipping over part of the gameplay.



it can be noticed that the time it takes for packets to be sent is not the same as the time in which they will arrive at the receiver side. One of the packets faces an unexpected delay on its way and is received after the expected time. **This is jitter**